



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Estimate the Percent of a Number

Lesson 4-3 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

Language Objective: I can explain an estimate using the words about, benchmark, and compatible numbers.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

estimate

benchmark percent

compatible numbers

double number line

the whole



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A reasoned answer close enough to be useful is an .

2

A friendly percent used as a landmark, like 10% or 25%, is a

3

Numbers close to the originals that are easy to compute with are

4

Two parallel lines pairing amounts with their percents form a

5

The quantity that counts as 100% in a problem is

 .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Of the USED storage, approximately what percent are photos — and why does the whole change from 6 GB to 4 GB for this question?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **estimate** — A reasoned answer that is close enough to be useful, found without exact computation.

estimación — Una respuesta razonada, lo bastante cercana para ser útil, sin cálculo exacto.

- ☐ **benchmark percent** — A friendly percent — 1%, 10%, 25%, 50%, 75%, 100% — used as a landmark for estimating.

porcentaje de referencia — Un porcentaje amigable — 1%, 10%, 25%, 50%, 75%, 100% — que sirve de punto de referencia.

- ☐ **compatible numbers** — Numbers close to the originals that are easy to compute with mentally.

números compatibles — Números cercanos a los originales con los que es fácil calcular mentalmente.

- ☐ **double number line** — Two parallel number lines that pair amounts with their percents at the same positions.

recta numérica doble — Dos rectas paralelas que emparejan cantidades con sus porcentajes en las mismas posiciones.

- ☐ **the whole** — The quantity that counts as 100% in a problem — and it can change between questions.

el todo — La cantidad que representa el 100% en un problema, y puede cambiar entre preguntas.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The whole for this question is ____ . Photos are about ____ % of the used storage, since 3.5 GB sits between the ____ % benchmark and the ____ % benchmark on the number line.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 4 GB sits between two benchmark landmarks on the number line, closer to the 75% mark, so a reasonable estimate lands near there without computing the exact fraction.

Evidence: A strong answer names both landmark percents (50% and 75%) and explains why 4 GB sits closer to one than the other. A weak answer just says 'it's close to 70%' without placing it between benchmarks.

Reasoning: This evidence connects the definition of estimate to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The whole for this question is ____ . Photos are about ____ % of the used storage, since 3.5 GB sits between the ____ % benchmark and the ____ % benchmark on the number line.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- I know because ____.
Lo sé porque ____.
- So ____.
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** estimate
2. **Notes 2:** benchmark percent
3. **Notes 3:** compatible numbers
4. **Notes 4:** double number line
5. **Notes 5:** the whole
6. **Watch for this mistake:** *Keeping the same whole for every question — when the problem says 'of the used storage,' the whole changes from 6 GB to 4 GB, and estimates anchored to the old whole are wrong before any arithmetic happens.*
7. **Try It 1:** A. About one third of 80, so about \$26 off — 32% is near the benchmark $\frac{1}{3}$, and \$79.95 is nearly 80. A third of 80 is about 26.7.
8. **Try It 2:**